



NEUTRON & X-RAY OPTICS

# Shaping, focusing, and timing the beam

Mirrors, lenses, crystals, multilayers and choppers in McStas & McXtrace

McStas — neutrons

McXtrace — X-rays



---

Peter Willendrup · DTU Physics

PaNRAID School · supported by PEPR DIADEM



# What we'll cover

Everything between source and sample that reshapes the beam in space, direction, energy or time.

1

## Five ways to shape a beam

A quick map of the optics families we'll cover

3

## Grazing-incidence reflection

The shared physics behind guides and mirrors

5

## Refractive lenses (CRLs)

Why X-ray and neutron lenses must be stacked

7

## Multilayers

One interference effect, two very different jobs

2

## Spatial & directional shaping

Slits, apertures and collimators

4

## Guides & mirrors

Channelling neutrons vs. focusing X-rays

6

## Crystal monochromators

Bragg's law, mosaicity, and geometry

8

## Temporal shaping

Choppers, velocity selectors, and why neutrons are special



# Five ways to shape a beam

Every optical component in either code falls into one of these families.

| Spatial                                                                                    | Reflective                                                       | Refractive                                                                          | Diffractive                                                                  | Temporal                                                           |
|--------------------------------------------------------------------------------------------|------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------|
| 1<br><br>Pinholes, slits, apertures, collimators — define where the beam is allowed to be. | 2<br><br>Mirrors & guides — steer or focus at grazing incidence. | 3<br><br>Compound lenses (CRLs) — bend the beam like an ordinary lens, just barely. | 4<br><br>Crystals & multilayers — select wavelength/energy via interference. | 5<br><br>Choppers & selectors — shape the beam in time, not space. |

*We'll move roughly in this order — spatial first, temporal last — flagging where neutrons and X-rays share a component and where they don't.*

# Spatial & directional shaping

## Spatial: pinholes, slits, apertures

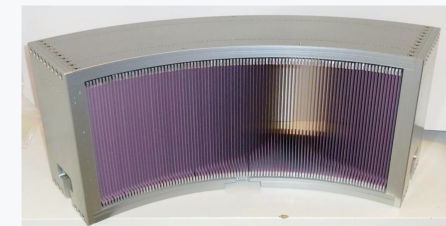


- A static or adjustable “hole” in absorbing material
- Defines the beam's spatial footprint downstream
- Same component idea in both codes: Slit

## Directional: collimators



- Two or more slits (or a channel array) define an accepted divergence
- Absorbing walls create parallel “channels” through the collimator
- Usually static, but rotating radial collimators exist — e.g. to reject sample-environment scattering

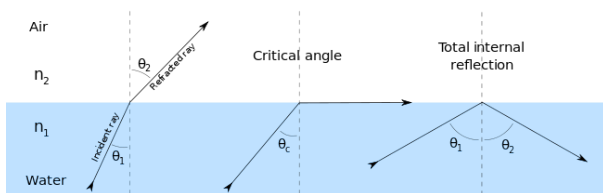


**Fully shared:** these are geometric, energy-independent devices — the same component names and logic apply whether the beam is neutrons or photons.

# Grazing-incidence reflection

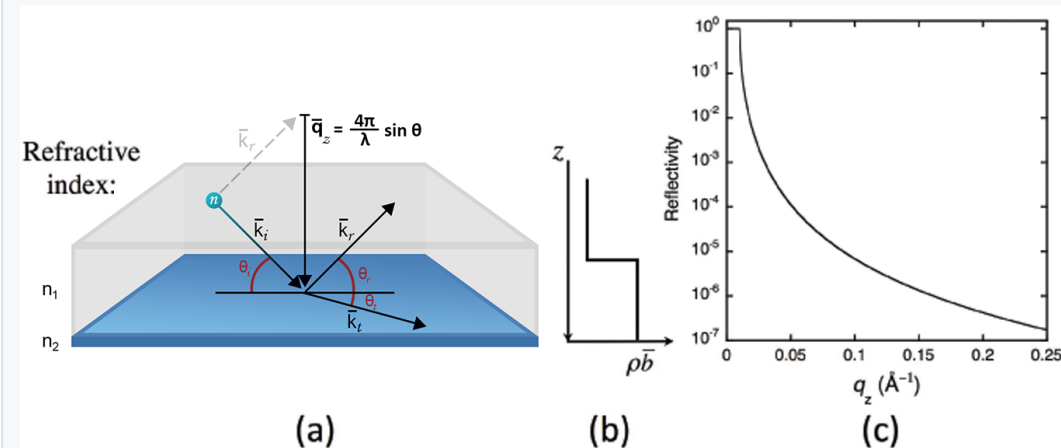
- For both neutrons and X-rays, matter's refractive index is just below 1:  $n = 1 - \delta (+ i\beta$  for absorption).
- Going from vacuum ( $n=1$ ) into a medium with  $n<1$  mirrors the classic total-internal-reflection picture — here it's called total external reflection.
- Below a small critical angle  $\theta_c$ , a ray hitting a smooth, flat surface reflects almost perfectly — no coating chemistry required, just the right angle.
- This single effect underlies both neutron guides and X-ray mirrors — what differs is how it's exploited.

Visible light – water/air interface:  
Total **internal** reflection



Neutrons and X-rays – vacuum/matter interface:  
Total **external** reflection

Below  $\theta_c$ : total reflection

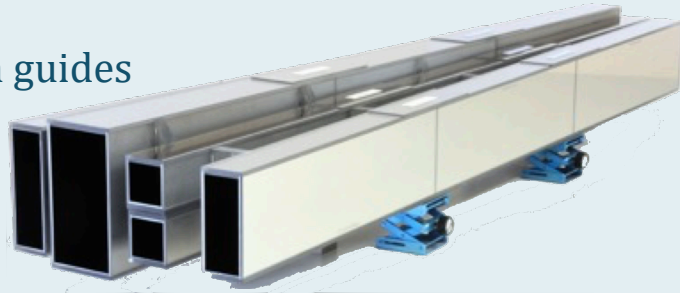


$\theta_c$  is tiny — typically a fraction of a degree — for both probes, which is why guides and mirrors are long, shallow, and need to be extremely flat/smooth.

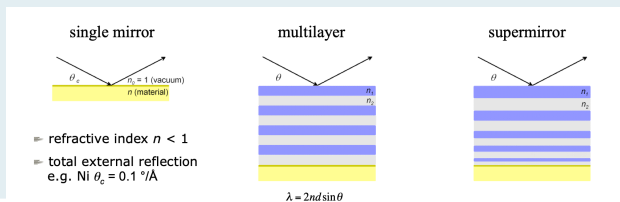
# Guides & mirrors

Same critical-angle physics, but neutron beamlines and X-ray beamlines put it to work differently.

## McStas · neutron guides

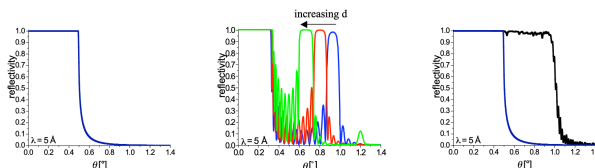


- Guide, Guide\_channeled, Guide\_gravity: long supermirror-coated channels
- Many shallow bounces over metres — preserves intensity over long transport distances
- Coating quality set by an “m-value” — multiples of the natural-Ni critical angle



## McXtrace · X-ray mirrors

- Mirror\_curved, TwinKB\_ML: one or two large, single-bounce focusing mirrors
- Typically one bounce per axis (e.g. a Kirkpatrick-Baez pair) rather than a many-bounce channel
- Reflectivity data files (measured or computed) describe the coating's angle/energy response



Why the difference? Neutron sources are weak per-neutron, so long guides trade length for preserved flux; X-ray sources are so bright that a couple of well-figured mirrors are enough to focus or steer the beam.

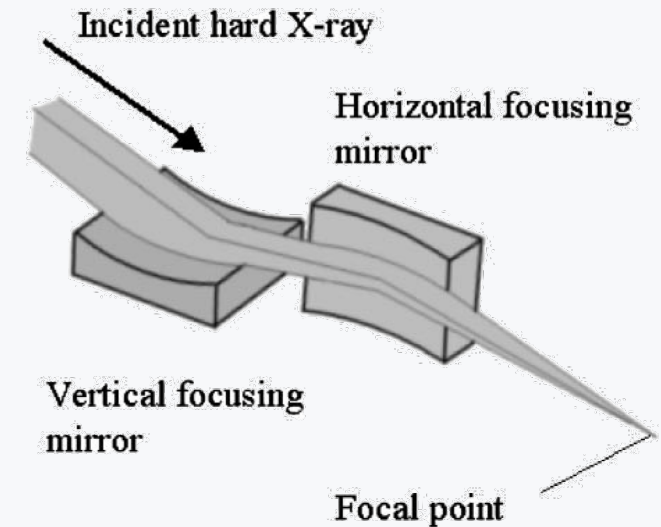
# KB mirrors, in practice

A Kirkpatrick-Baez (KB) pair: two perpendicular curved mirrors, one per focusing axis.

- First (meridional) mirror focuses one axis; a glancing angle  $\theta$  redirects the beam by  $2\theta$ .
- An Arm component re-establishes the optical axis after each reflection — it does nothing physically, it just carries a rotated coordinate frame forward.
- The second mirror sits on an Arm rotated  $90^\circ$  around the beam axis, so it focuses the perpendicular direction.
- Focal lengths follow directly from the glancing angle and radius of curvature  $R$ :

$$\begin{aligned} f_m &= R \cdot \sin(\theta/2) && \text{(meridional / 1st mirror)} \\ f_s &= R/2 \cdot \sin(\theta) && \text{(sagittal / 2nd mirror)} \end{aligned}$$

source → mirror 1 → mirror 2 → focus



$\vartheta$  is typically only a few mrad — chosen from the material's critical angle at the working energy.

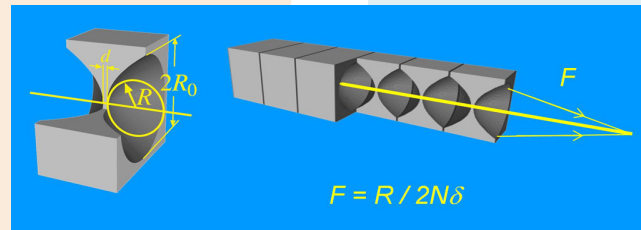
# Refractive lenses (CRLs)

Both codes offer compound refractive lenses — and both need them for the same reason.

## X-ray

$$\delta_{\text{Xray}} = f(Z, \lambda)$$

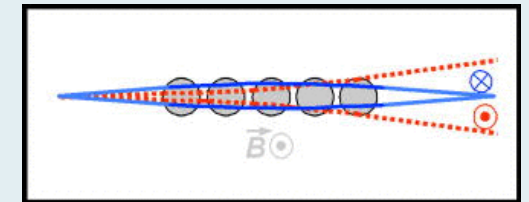
- Grows with electron density — higher atomic number  $Z$  means more refraction, but also more absorption
- Low- $Z$ , low-absorption materials preferred: Be, Al, polymers



## Neutron

$$n = 1 - \delta_{\text{neutron}} = f(b, B)$$

- Governed by coherent scattering length  $b$ , plus a magnetic term  $B$
- The magnetic (spin-dependent) lensing effect is small in practice — hence ordinary CRLs are still the practical choice



**Why “compound”:** in both regimes the index is barely below 1, so a single lens bends the beam almost imperceptibly. Stacking many (parabolic) lenses in series — a compound refractive lens — adds up to a usable focal length.

# A CRL beamline, worked example

| Stack   | Material | R (tip)           | Lenses | Diameter | Thickness (tip)  |
|---------|----------|-------------------|--------|----------|------------------|
| Stack 1 | Be       | 200 $\mu\text{m}$ | 16     | 1 mm     | 50 $\mu\text{m}$ |
| Stack 2 | Al       | 200 $\mu\text{m}$ | 10     | 1 mm     | 20 $\mu\text{m}$ |

**Setup:** Source\_gaussian at  $E_0=23.32$  keV ( $dE=1$  keV), CRL stack 31.5 m downstream, focused to a spot 10 m further on. A 2D + energy monitor at the stack checks the incoming footprint and spectrum; another monitor is scanned in  $z$  to find the true focus.

- Directional sampling (focus\_xw, focus\_yh, dist on the source) concentrates statistics on the aperture that matters, without biasing intensity.
- Used together with a slit, the CRL's achromatic focusing can double as a crude, high-throughput monochromator.
- mxdisplay ("Trace" mode) lets you visually follow rays through the lens stack — useful for sanity-checking alignment.
- Swapping Be  $\leftrightarrow$  Al lets you compare absorption vs. refraction trade-offs directly.



# Crystal monochromators: Bragg's law

`n λ = 2 d sin(θ)`

— exactly the same equation, in both codes

|                 | X-ray                                                                    | Neutron                                                           |
|-----------------|--------------------------------------------------------------------------|-------------------------------------------------------------------|
| Typical crystal | Perfect single crystal: Si, Ge, C                                        | Mosaic crystal: pyrolytic graphite, deformed Cu, Ge               |
| Mosaicity       | Very low                                                                 | Often high (multi-crystal, focusing arrays)                       |
| Geometry        | Laue (transmission) or Bragg (reflection)                                | Almost always Bragg (reflection)                                  |
| Resulting beam  | Narrow, low divergence and $\frac{\delta\lambda}{\lambda} \approx 0.1\%$ | Medium divergence and $\frac{\delta\lambda}{\lambda} \approx 3\%$ |



In both codes: Bragg\_crystal (McXtrace) and Monochromator\_flat / Monochromator\_curved / Single\_crystal (McStas) all rotate to satisfy Bragg's law — the rotation angle is a natural scan parameter for a rocking curve.

# Bent crystals & double-crystal monochromators

## Bending a crystal

- A radius of curvature on the crystal (Bragg\_crystal\_bent) is a simple way to model a heat-bump — thermal distortion from absorbed beam power.
- Compare a Divergence\_monitor before/after bending to see the extra divergence introduced.
- A rocking curve (scanning  $\theta$ ) shows the peak shape change directly.

## Double-crystal monochromators (DCM)

- A second crystal, rotated by the negative of the first angle, brings the beam back parallel to the incoming direction
- Arms track the beam axis between crystals; the pair can share a single rotation stage, as in a real DCM
- Laue\_crystal\_BC swaps reflection for transmission geometry — useful for higher-energy or higher-heat-load beamlines

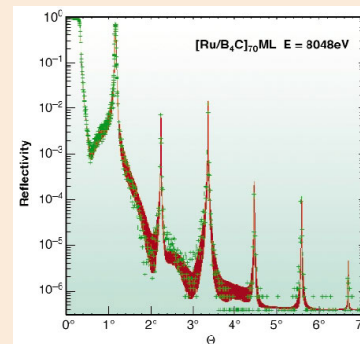
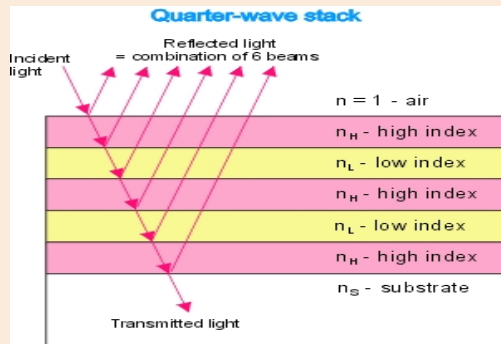
```
// crystal 2, mirrored relative to crystal 1's Arm
COMPONENT xtal2 = Bragg_crystal(...)
AT (0,0,0) RELATIVE arm_after_xtal1
ROTATED (-MONO_ANGLE, 0, 0) RELATIVE arm_after_xtal1
```

# Multilayers: one concept, two jobs

A stack of thin alternating layers reflects by interference — like Bragg's law, but for a synthetic superperiod.

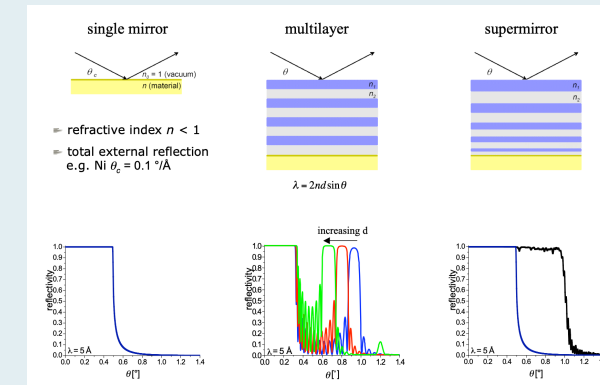
## X-ray: a broadband monochromator

- Used as a monochromator with a much broader bandpass — and higher flux — than a perfect crystal
- Pick one of the first “side peaks” on the reflectivity curve
- Curved multilayers combine monochromation with focusing



## Neutron: a “supermirror” coating

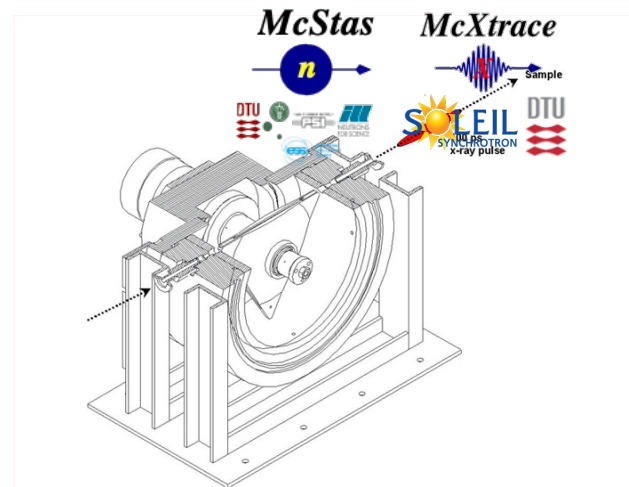
- Stacked Bragg-like reflections extend the total-external-reflection range well beyond a single critical angle
- Ideally a wide, flat reflectivity vs. Q
- Used to coat neutron guides — effectively an “optical fibre” for neutrons



Same underlying phase-interference physics in both codes — only the intended job (monochromate vs. transport) differs.

# Temporal shaping: choppers

- A rotating absorbing disk with one or more slits cuts a continuous beam into pulses in time.
- Both X-ray and neutron beamlines use a disk-chopper-like component to define or measure time structure — e.g. for time-of-flight (TOF) measurements.
- The physical picture is identical: a slit sweeps across the beam at a set frequency, and only rays arriving within the open window get through.



**Where it stops being shared:** only neutrons are slow enough that a mechanical rotor can also act as an energy filter. An X-ray chopper shapes time only — it can't pick out an energy band the way a neutron velocity selector or Fermi chopper can.

*Up next: two neutron-only rotating devices that filter on speed, not just on time.*



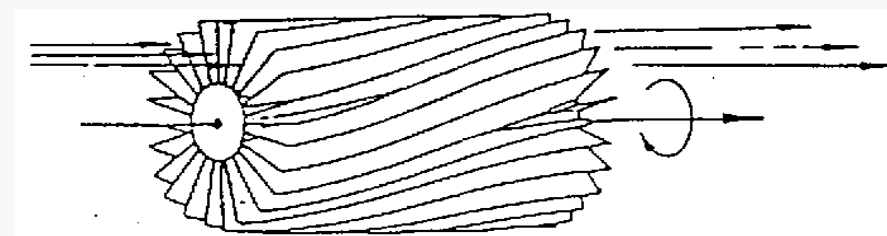
# Velocity selectors

A rotating drum of twisted, absorbing blades — only a narrow velocity (wavelength) band makes it straight through.

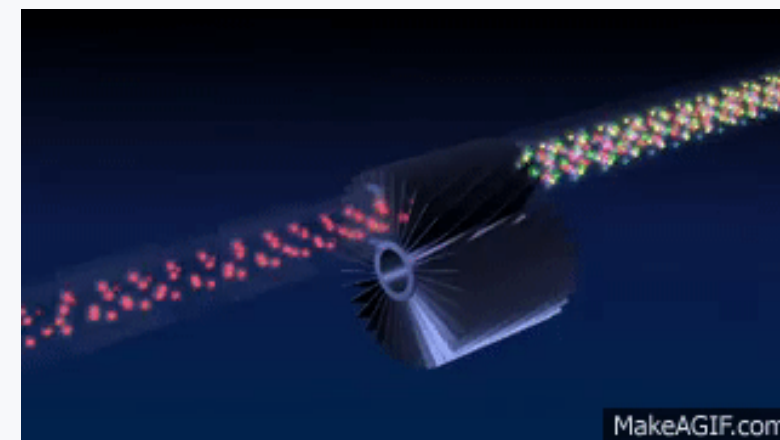


| Parameter           | Meaning                                  |
|---------------------|------------------------------------------|
| xwidth, yheight [m] | entry aperture                           |
| zdepth [m]          | housing length                           |
| length, d [m]       | blade length / thickness                 |
| alpha [deg]         | blade twisting angle                     |
| radius [m]          | rotation axis → aperture-centre distance |
| nu [Hz]             | rotation speed (counterclockwise)        |
| nslit               | number of blades                         |

twisted absorbing blades



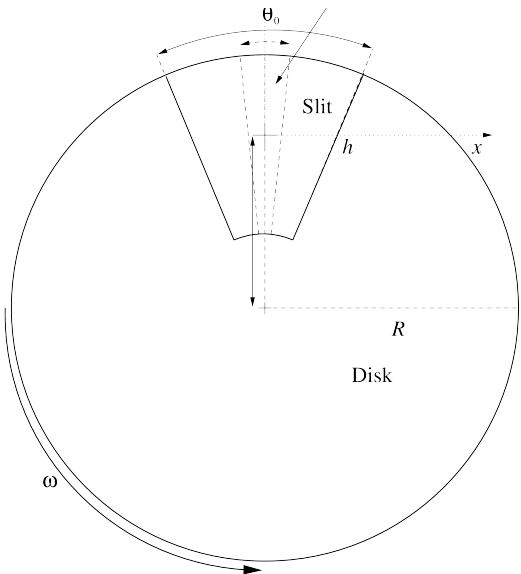
broad monochromatization  
 $\delta\lambda/\lambda \approx 10\%$



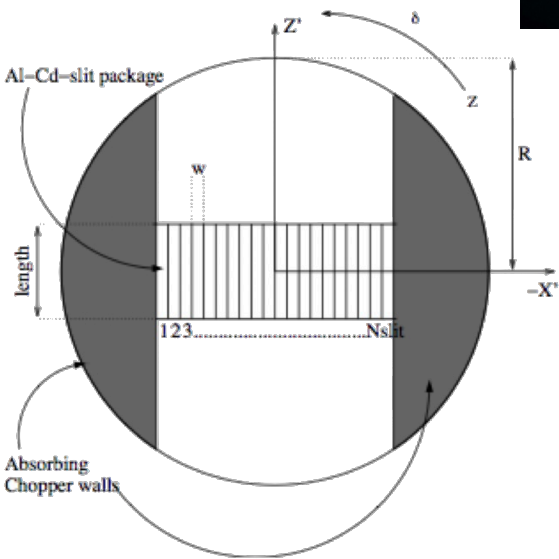
# Disk & Fermi choppers



## DiskChopper



## Fermi chopper



- A small drum of curved slit packages, spinning fast, sitting directly in the beam — a “rotating collimator”.
- Only neutrons with the right speed can cross the curved channel before it rotates shut — so it acts as a combined fast time-gate and coarse energy filter.
- Used right next to the sample, where a disk chopper's timing alone is no longer selective enough.

**Both are McStas-only:** there's no X-ray analogue — the mechanical rotation speeds needed would be absurd at the speed of light. X-ray timing instead comes from the accelerator's bunch structure itself.

| Parameter                            | Meaning                                              |
|--------------------------------------|------------------------------------------------------|
| <code>nu, theta_0</code>             | frequency; angular slit width                        |
| <code>radius, yheight, xwidth</code> | disk geometry / slit opening                         |
| <code>jitter, delay, phase</code>    | timing: random jitter, fixed delay, or angular phase |
| <code>isfirst, npulse</code>         | marks the first chopper; pulses to generate if so    |



# One optics toolkit, two beams

## What's shared

- Spatial shaping: slits & collimators, same logic either way
- Grazing-incidence total external reflection ( $n < 1$ , critical angle)
- Bragg's law, and the idea of rotating a crystal to tune it
- Refractive-index-just-below-1 logic behind compound lenses
- Multilayer interference physics
- The chopper concept for defining time structure

## What differs

- Many-bounce guides (neutron) vs. one/two-bounce focusing mirrors (X-ray)
- Crystal choice: near-perfect low-mosaicity vs. mosaic/multi-crystal
- Multilayer role: X-ray monochromator vs. neutron guide coating
- Velocity selectors & Fermi choppers exist only for neutrons
- X-ray timing comes from the accelerator, not a mechanical rotor

# Where to go deeper

## McStas

`mcdoc optics / mcdoc -c`

Guide, Guide\_channeled, Monochromator\_flat/curved, DiskChopper, FermiChopper, V\_selector

[mcstas.org](http://mcstas.org)

## McXtrace

`mxdoc optics / mxdoc -c`

Practical exercises: CRLs, KB mirrors, monochromators — pick two of the three

[mcxtrace.org](http://mcxtrace.org)

---

Peter Willendrup · [pkwi@fysik.dtu.dk](mailto:pkwi@fysik.dtu.dk) · DTU Physics